Ana Sofía M. Uzsoy

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My research interests include large survey science, galaxy evolution, Lyman-Alpha Emitter galaxies, galaxy-environment interactions, and applications of statistical and machine learning for astrophysics.

Education

Harvard University
Cambridge, MA, USA

PhD, Astronomy & Astrophysics (in progress) *September 2022 - May 2027 (exp.)*

- Advisor: Doug Finkbeiner

University of Cambridge
Cambridge, UK

M.Phil, Machine Learning and Machine Intelligence *September 2021 - August 2022*

- Advisor: Kaisey Mandel
- Thesis: *Scalable Bayesian Inference for Probabilistic Spectrotemporal Models of Type Ia Supernovae*

North Carolina State University (NCSU)
Raleigh, NC, USA

B.S. Computer Science, B.S. Physics *August 2017 - May 2021*

- *Summa cum laude*, Dean's List (All Semesters)
- Minors: Mathematics, Oboe Performance

Work Experience

Google
Santa Barbara, CA, USA (remote)

Software Engineering Intern, Quantum AI team *June 2021 – August 2021*

- Wrote new methods to improve quantum computing simulators in Cirq (Google's quantum computing library) by modeling noise from quantum hardware.
- Wrote internal software design documents and public documentation for Cirq.

Google
Mountain View, CA, USA (remote)

Software Engineering Intern, Tensorflow/Keras/Kaggle teams *May 2020 – July 2020*

- Created natural language processing (NLP) machine learning code examples using TensorFlow 2 for Kaggle, Google's online machine learning and data science competition platform.
- Designed and led launch of new NLP-focused Kaggle competition that remains active.
- Wrote text generation code featured in Google's Data Analyst Professional Certificate Course.

NASA Langley Research Center
Hampton, VA, USA

Intern *June 2019 – August 2019*

- Worked on team for SAGE III ISS (Stratospheric Aerosol & Gas Experiment III on the International Space Station), an atmospheric science experimental apparatus aboard the ISS.
- Wrote data analysis pipeline to identify cloud interference in spectroscopic atmosphere measurements using neural networks with PyTorch.

Achievements

2026 **Young Alumni "Rising Star" Award**, NCSU Computer Science Department

2026 **Astrostatistics Student Paper Competition Finalist**, American Statistical Association

2021 **National Science Foundation Graduate Research Fellowship**, awarded to outstanding graduate students in STEM disciplines at US institutions

2021 **Churchill Scholarship**, national scholarship providing funding for American students in STEM disciplines to complete an MPhil at the University of Cambridge

2021 **Rodney I. McCormick Award**, awarded by NCSU Physics Department to a graduating senior for excellence in undergraduate research

2020 **Phi Beta Kappa**, inducted into national honor society for high-achieving undergraduates

2020 **First place poster prize**, McCormick Research Symposium, NCSU Physics Department

2020 **NCSU Computer Science Department Faculty Senior Scholar**, awarded to a rising senior based on academic excellence, intellectual breadth, and depth of character

2019 **Barry M. Goldwater Scholarship**, a national scholarship awarded to promising undergraduates who plan to pursue a STEM research career

2017 **Park Scholarship**, a full-ride merit scholarship to NCSU awarded on the basis of outstanding accomplishments and potential in scholarship, leadership, character, and service

Publications (with links)

First- and Second-authored papers

Uzsoy, **A. S. M.**, Lamman, C., Zhu, P., & Weber, M. (2026). Tracing the evolution of galaxy environments with manifold learning. https://arxiv.org/abs/2610.03850 [arxiv preprint] (submitted to *Physical Review X Intelligence*)

Uzsoy, **A. S. M.**, & Villar, V. A. (2026). Photometry is all you need: supernova classification as a mixing problem. [arxiv preprint] (submitted to *The Astrophysical Journal Supplement Series*)

Uzsoy, **A. S. M.**, Saydjari, A. K., Dey, A., Raichoor, A., Finkbeiner, D. P., et al. (2026). Bayesian Component Separation for DESI LAE Automated Spectroscopic Redshifts and Photometric Targeting. *The Astrophysical Journal*, *999*(2), 241. [Journal Article]

Uzsoy, **A. S. M.**, Dey, A., Raichoor, A., Finkbeiner, D. P., Ramakrishnan, V., et al. (2025). Effect of local environment on Lyα line profile in DESI/ODIN LAEs. [arxiv preprint] (accepted by *The Astrophysical Journal*)

Uzsoy, **A. S. M.**, Lamman, C., & Weber, M. (2025). Manifold Learning for Cosmic Structures. *Machine Learning and the Physical Sciences Workshop @ NeurIPS 2025*. [ML workshop paper]

de Soto, K., **Uzsoy**, **A. S.**, & Villar, V. A. (2024). Sharing Space: A Survey-agnostic Variational Autoencoder for Supernova Science. *Machine Learning and the Physical Sciences Workshop @ NeurIPS 2024*. [ML workshop paper]

Uzsoy, **A. S. M.**, Thorp, S., Grayling, M., & Mandel, K. S. (2024). Variational inference for acceleration of SN Ia photometric distance estimation with BayeSN. *Monthly Notices of the Royal Astronomical Society*, *535*(3), 2306–2321. [Journal Article]

Saydjari, A. K., **Uzsoy**, **A. S. M.**, Zucker, C., Peek, J. E. G., & Finkbeiner, D. P. (2023). Measuring the 8621 Å Diffuse Interstellar Band in Gaia DR3 RVS Spectra: Obtaining a Clean Catalog by Marginalizing over Stellar Types. *The Astrophysical Journal*, *954*(2), 141. [Journal Article]

Uzsoy, **A. S. M.**, Zareiesfandabadi, P., Jennings, J., Kemper, A. F., & Elting, M. W. (2021). Automated tracking of *S. pombe* spindle elongation dynamics. *Journal of Microscopy*, *284*(1), 83–94. [Journal Article]

Uzsoy, **A. S. M.**, Rogers, L. A., & Price, E. M. (2021). Radius and Mass Distribution of Ultra-short-period Planets. *The Astrophysical Journal*, *919*(1), 26. [Journal Article]

Co-authored papers

SDSS Collaboration, Aghakhanloo, M., Aguilar, D., Aird, J., & et al [incl. **A. S. M. Uzsoy**]. (2026). The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper. [arxiv preprint] (submitted to *The Astrophysical Journal Supplement Series*)

Ortiz, A., Ramakrishnan, V., Lee, K.-S., Dey, A., & et al [incl. **A. S. M. Uzsoy**]. (2026). ODIN: Confirmation and 3D Reconstruction of Six Massive Protoclusters at Cosmic Noon. *The Astrophysical Journal*, *1008*(1), 85. [Journal Article]

Grayling, M., Thorp, S., Mandel, K. S., Dhawan, S., **Uzsoy**, **A. S. M.**, et al. (2024). Scalable hierarchical BayeSN inference: investigating dependence of SN Ia host galaxy dust properties on stellar mass and redshift. *Monthly Notices of the Royal Astronomical Society*, *531*(1), 953–976. [Journal Article]

Presentations

Talks

CCAPP Seminar, The Ohio State University; Columbus, OH	*September 29 2026*
Observer Meeting, Northwestern University; Evanston, IL	*September 25 2026*
University of Chicago; Chicago, IL	*September 24 2026*
NSF IAIFI Summer Workshop, MIT; Cambridge, MA	*August 11 2026*
Joint Statistical Meeting; Boston, MA	*August 2 2026*
University of Edinburgh; Edinburgh, Scotland	*July 16 2026*
DESI Collaboration Meeting; Durham, England	*July 13 2026*
University of Cambridge Astro Data Science Seminar; Cambridge, England	*July 9 2026*
AstroAI Workshop, Center for Astrophysics | Harvard & Smithsonian; Cambridge, MA	*June 17 2026*
Galaxy Formation Group Meeting, UC Berkeley; Berkeley, CA	*March 18 2026*
Stanford KIPAC Visitor Tea; Palo Alto, CA	*March 13 2026*
DESI Lunch, Lawrence Berkeley National Lab; Berkeley, CA	*March 11 2026*
ITC Luncheon, Center for Astrophysics | Harvard & Smithsonian; Cambridge, MA	*February 12 2026*
DESI Collaboration Meeting; Tucson, AZ	*December 11 2025*
AstroAI Seminar, Center for Astrophysics | Harvard & Smithsonian; Cambridge, MA	*November 3 2025*

University of Toronto/CITA Cosmology Lunch; Toronto, Canada *October 10 2025*
UT Austin Galaxies & Cosmology Seminar; Austin, TX *October 6 2025*
Open SkAI Workshop; Chicago, IL *September 5 2025*
University of Cambridge Astro Data Science Seminar; Cambridge, England *July 21 2025*
AstroAI Seminar, Center for Astrophysics | Harvard & Smithsonian; Cambridge, MA *May 19 2025*
Universidad de La Serena; La Serena, Chile *April 24 2025*
Escape of Lyman Radiation from Galactic Labyrinths; Crete, Greece *April 11 2025*
Princeton Astro Machine Learning Journal Club; Princeton, NJ *February 6 2025*
Transient Tea, Center for Astrophysics | Harvard & Smithsonian; Cambridge, MA *January 24 2025*
DESI Lunch, Lawrence Berkeley National Lab; Berkeley, CA *October 9 2024*
Stanford KIPAC Visitor Tea; Palo Alto, CA *October 8 2024*
Astromerique Seminar, U. of Montreal Ciela Institute; Montreal, Canada (virtual) *October 1 2024*
NSF IAIFI Summer Workshop, MIT; Cambridge, MA *August 13 2024*
DESI Collaboration Meeting; Marseille, France *July 12 2024*
NSF's NOIRLab; Tucson, AZ *May 24 2024*
Harvard CHASC Astrostatistics Seminar; Cambridge, MA *February 7 2024*
NCSU Physics Department McCormick Symposium; Raleigh, NC (virtual) *May 5 2021*
University of Chicago STEM Research Symposium; Chicago, IL *August 1 2018*
Leadership Alliance National Symposium; Hartford, CT *July 28 2018*

Posters

European Astronomical Society Meeting; Lausanne, Switzerland *June 29-July 3, 2026*
NeurIPS Machine Learning and the Physical Sciences Workshop; San Diego, CA *December 6 2025*
DESI Collaboration Meeting; Berkeley, CA *July 7–11 2025*
Rare Gems in Big Data; Tucson, AZ *May 20–24 2024*
243rd American Astronomical Society Meeting; New Orleans, LA *January 7–12 2024*
NCSU Physics Department McCormick Symposium; Raleigh, NC (virtual) *May 6 2020*
235th American Astronomical Society Meeting; Honolulu, HI *January 4–8 2020*
2019 American Society for Cell Biology Meeting; Washington, DC *December 7–11 2019*
Triangle Cytoskeleton Meeting; Saxapahaw, NC *September 30 2019*
NCSU Spring Undergraduate Research Symposium; Raleigh, NC *April 24 2019*
NCSU Physics Department McCormick Symposium; Raleigh, NC *April 22 2019*
Triangle Cytoskeleton Meeting; Saxapahaw, NC *September 24 2018*
Intel International Science & Engineering Fair; Los Angeles, CA *May 14–19 2017*

Mentorship

Helena Brittain *August 2026 - present*

- Harvard undergraduate, working on senior thesis on Bayesian component separation applications for photometry

Rong Xu *September 2026 - present*

- Harvard undergraduate, working on junior thesis on data-driven priors for Lyman-Break galaxies in DESI

Teaching & Service

Paper Referee *June 2025 - present*

- I have served as a reviewer for AI + science workshops at machine learning conferences NeurIPS, ICML and ICLR, and also for DESI Collaboration-Wide Revew.

Leadership Team, AstroAI *February 2025 - present*

- Serve on committees and help organize talks and workshops for AstroAI, a center dedicated to interdisciplinary research at the intersection of artificial intelligence and astrophysics at the Center for Astrophysics | Harvard & Smithsonian.

Derek Bok Center Teaching Certificate *July 2025*

- Received a teaching certificate from the Derek Bok Center for Teaching and Learning at Harvard. Requirements included participating in three pedagogy seminars, receiving teaching feedback via a classroom video consultation, creating an original syllabus, and writing a teaching statement.

Teaching Fellow, Machine Learning for Astrophysics *Spring 2024, 2025*

- Guest lectured, held office hours, graded assignments, provided feedback, and supported students in Astro 205, a graduate course on machine learning techniques for astrophysics at Harvard.

Scientific/Local Organizing Committee, AstroAI Workshop *February 2024 - June 2024*

- Helped arrange speakers, coordinated volunteers, reviewed abstracts, and chaired sessions for the 2024 AstroAI Workshop at the Center for Astrophysics | Harvard & Smithsonian.

President, NCSU Women in Physics *August 2020-May 2021*

- Organized a peer-mentoring program, speaker series and social events to promote a community of women and gender minorities in the NCSU physics department.

NCSU College of Sciences Student Ambassador *August 2018-May 2021*

- Represented the NCSU College of Sciences and physics department at university events for current and prospective students.